

Spiny-rayed fishes

PHYLUM	Chordata
CLASS	Osteichthyes
SUBCLASS	Actinopterygii
SUPERORDER	Acanthopterygii
ORDERS	34
FAMILIES	284
SPECIES	About 14,800

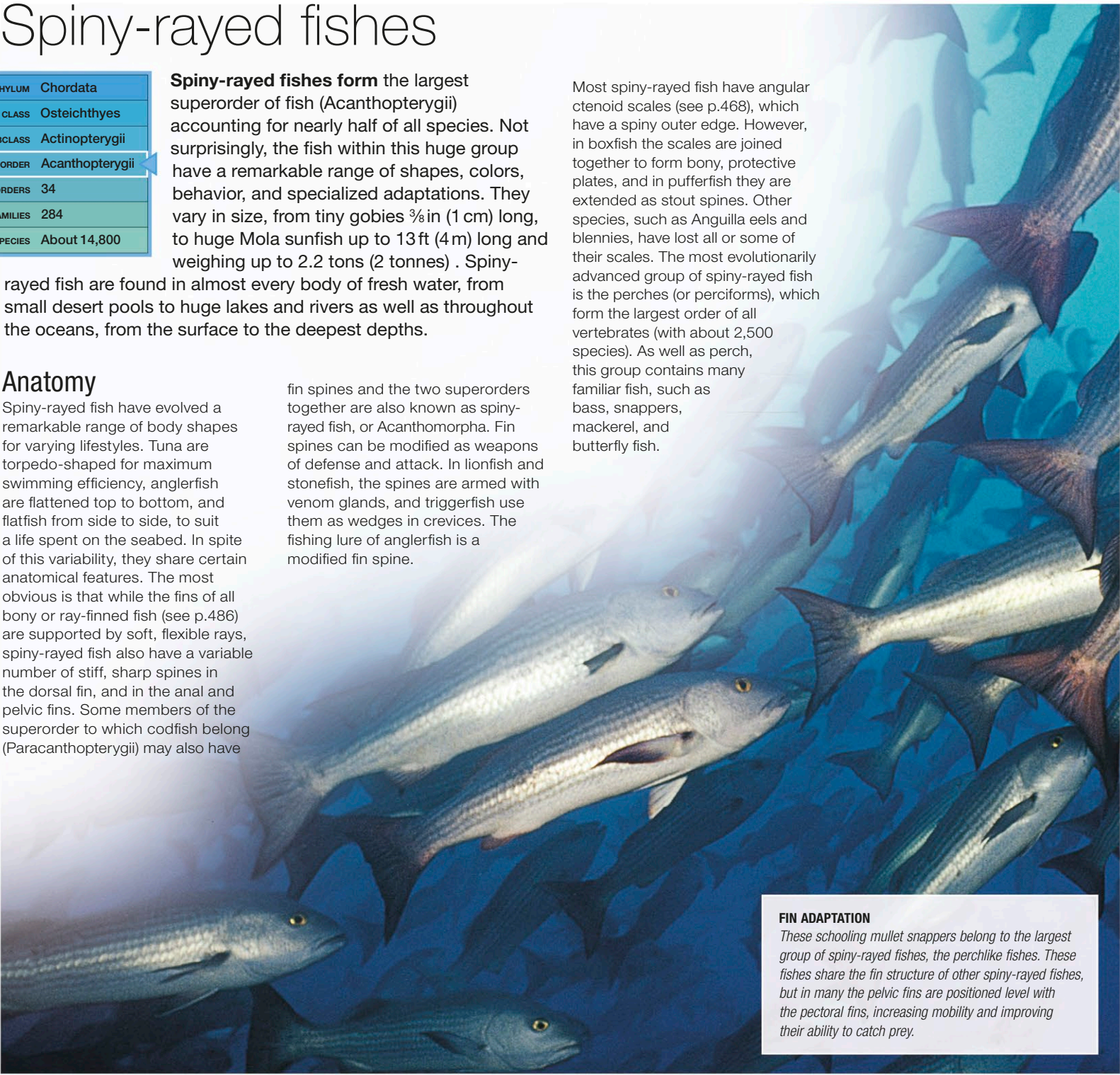
Spiny-rayed fishes form the largest superorder of fish (Acanthopterygii) accounting for nearly half of all species. Not surprisingly, the fish within this huge group have a remarkable range of shapes, colors, behavior, and specialized adaptations. They vary in size, from tiny gobies $\frac{3}{8}$ in (1 cm) long, to huge Mola sunfish up to 13 ft (4 m) long and weighing up to 2.2 tons (2 tonnes) . Spiny-rayed fish are found in almost every body of fresh water, from small desert pools to huge lakes and rivers as well as throughout the oceans, from the surface to the deepest depths.

Anatomy

Spiny-rayed fish have evolved a remarkable range of body shapes for varying lifestyles. Tuna are torpedo-shaped for maximum swimming efficiency, anglerfish are flattened top to bottom, and flatfish from side to side, to suit a life spent on the seabed. In spite of this variability, they share certain anatomical features. The most obvious is that while the fins of all bony or ray-finned fish (see p.486) are supported by soft, flexible rays, spiny-rayed fish also have a variable number of stiff, sharp spines in the dorsal fin, and in the anal and pelvic fins. Some members of the superorder to which codfish belong (Paracanthopterygii) may also have

fin spines and the two superorders together are also known as spiny-rayed fish, or Acanthomorpha. Fin spines can be modified as weapons of defense and attack. In lionfish and stonefish, the spines are armed with venom glands, and triggerfish use them as wedges in crevices. The fishing lure of anglerfish is a modified fin spine.

Most spiny-rayed fish have angular ctenoid scales (see p.468), which have a spiny outer edge. However, in boxfish the scales are joined together to form bony, protective plates, and in pufferfish they are extended as stout spines. Other species, such as Anguilla eels and blennies, have lost all or some of their scales. The most evolutionarily advanced group of spiny-rayed fish is the perches (or perciforms), which form the largest order of all vertebrates (with about 2,500 species). As well as perch, this group contains many familiar fish, such as bass, snappers, mackerel, and butterfly fish.

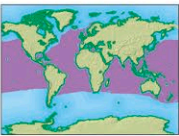


FIN ADAPTATION

These schooling mullet snappers belong to the largest group of spiny-rayed fishes, the perchlike fishes. These fishes share the fin structure of other spiny-rayed fishes, but in many the pelvic fins are positioned level with the pectoral fins, increasing mobility and improving their ability to catch prey.

Regalecus glesne

Oarfish



Length Up to 26 ft (8 m)
Weight Not recorded
Sex Male/Female
Status Least concern

Location Tropical, subtropical, and temperate waters worldwide



Although it lives worldwide in tropical and temperate seas, this huge, long, ribbonlike deep-sea fish is rarely captured, or even seen alive. Hence little is known about its behavior. Its

extraordinary length protects it against most predators and has made it the subject of stories involving sea serpents and marine "monsters." This fish has a silvery body with oblique dusky bars, a short, bluish head, and deep red fins. Its dorsal fin runs along almost the full length of its body, and has long rays forming a crest over the fish's head. The pelvic fins are attached just behind its gill cover, and each consists of a long, single ray, with an expanded tip that looks like the blade of an oar, which is how the oarfish gets its name. Oarfish, along with opahs, may belong in their own superorder of spiny-rayed fish, called Lamprimorpha.

